

④

4. In the circuit shown below, a voltage source provides a sinusoidal voltage to three loads as shown. This system is in steady state, and operates at a frequency of 60[Hz]. It is known that:

+1 Load 1 absorbs 600[W] and delivers 500[VAR] = 600[VA] - 500j

+2 Load 2 absorbs 750[VA] with a leading power factor of 0.7. 750[VA] < -45.573°

+3 Load 3 absorbs 800∠15°[VA].

Find $i_x(t)$ for this system.

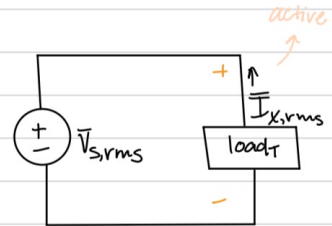
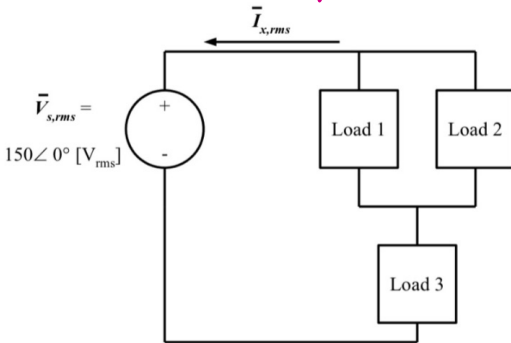
777.72 + 207.04j
degree will be negative

Reactive

$$\omega = 60[\text{Hz}] = 120\pi[\text{rad/sec}]$$

$$S_{L2} = \cos^{-1}(0.7) \quad 525 - 535 \cdot 20j$$

$$= 45.573^\circ$$



+2
redraw

$$S_{\text{del. by source}} = S_{\text{abs. by load 1}} + S_{\text{abs. by load 2}} + S_{\text{abs. by load 3}}$$

$$S_{\text{del. by source}} = (600[VA] - 500j) + (750[VA] \angle -45.573^\circ) + (800[VA] \angle 15^\circ)$$

$$S_{\text{del. by source}} = 2070.73[VA] \angle -23.586^\circ$$

$$2070.73[VA] \angle -23.586^\circ = -V_{s,rms} \cdot \bar{I}_{x,rms}^*$$

$$\bar{I}_{x,rms}^* = - \frac{2070.73[VA] \angle -23.586^\circ}{150[V_{rms}] \angle 0^\circ}$$

$$130[V_{rms}] \angle 0^\circ$$

$$\bar{I}_{x,rms}^* = 13.8049[A_{rms}] \angle 156.414^\circ$$

$$\bar{I}_{x,rms} = 13.8049[A_{rms}] \angle -156.414^\circ$$

Because it's in rms we want the original value

$$\text{so, } I_{rms} = \frac{I_x}{\sqrt{2}} \Rightarrow 13.8049[A_{rms}] (\sqrt{2}) = I_x$$

$$i_x(t) = 19.5231[A] \cos(120\pi[\text{rad/sec}]t - 156.414^\circ)$$

(+3 for trying)

* No penalty for
redraw if final
answer is correct